

G11 T1 C7 – Sample Mean, Variance, and Standard Deviation

Required equations. For a sample $x_1, x_2, \dots, x_n$, use the sample statistics

$$n = \text{number of observations}, \quad \bar{x} = \frac{\sum_{i=1}^n x_i}{n}, \quad s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}, \quad s = \sqrt{s^2}$$

Corresponding theoretical (population) values:

$$\mu = E(X), \quad \sigma^2 = \text{Var}(X), \quad \sigma = \sqrt{\sigma^2}$$

For a value–frequency table

$$n = \sum f, \quad \bar{x} = \frac{\sum fx}{n}, \quad s^2 = \frac{\sum f(x - \bar{x})^2}{n-1}.$$

Instructions. For every sample, calculate n , $\bar{x}$, s^2 , and s . Then compare each statistic with the given theoretical value μ , σ^2 , or σ . Round decimal answers to three decimal places.

1. **Battery operating time (individual data).** The operating times, in hours, of five randomly selected batteries are

8, 9, 10, 11, 12.

The manufacturer reports $\mu = 10$ hours and $\sigma^2 = 4$ hours².

2. **Local delivery time (individual data).** Six randomly selected delivery times, in minutes, are

18, 20, 21, 19, 22, 20.

Historical records give $\mu = 20$ minutes and $\sigma = 1.5$ minutes.

3. **Reusable cups (value–frequency table).** The number of reusable cups brought by each student in a random sample is summarized below. The school-wide model has $\mu = 2.5$ cups and $\sigma^2 = 1$ cup².

x	1	2	3	4
f	2	5	4	1

4. **Mobile-data use (value–frequency table).** Daily mobile-data use, in hundreds of megabytes, was recorded for ten students. The theoretical values are $\mu = 52$, $\sigma^2 = 4$, and $\sigma = 2$.

x	48	50	52	54
f	1	3	4	2

5. **Commute time (frequency buckets).** A random sample of commute times is grouped into the following intervals. Use class midpoints. The city model gives $\mu = 30$ minutes and $\sigma = 10$ minutes.

Time (min)	[10, 20)	[20, 30)	[30, 40)	[40, 50)
f	2	5	4	1

6. **Parcel mass (frequency buckets).** The masses of ten parcels are grouped below. Use class midpoints. Long-term records give $\mu = 4$ kg and $\sigma^2 = 3$ kg².

Mass (kg)	[0, 2)	[2, 4)	[4, 6)	[6, 8)
f	1	4	4	1

7. **Package preparation time (mixed data).** A random sample of package preparation times, in minutes, was recorded in two different ways. Three observations were recorded individually:

8, 10, 12.

The remaining observations were summarized in a value–frequency table:

x	9	10	11
f	2	3	2

Treat all the observations as one sample. Historical records give $\mu = 10$ minutes and $\sigma^2 = 2$ minutes².

8. **Customer service time (mixed data).** A random sample of customer service times, in minutes, was recorded using two different formats. Four observations were recorded individually:

22, 28, 32, 38.

The remaining observations were grouped as follows:

Time (min)	[10, 20)	[20, 30)	[30, 40)	[40, 50)
f	1	2	2	1

Treat all the observations as one sample and use class midpoints for the grouped observations. The theoretical values are $\mu = 30$ minutes and $\sigma = 9$ minutes.

1 Solutions

C7.1 – Battery operating time

$$n = 5, \quad \bar{x} = \frac{8 + 9 + 10 + 11 + 12}{5} = 10.$$

The sum of squared deviations is

$$\sum (x - \bar{x})^2 = 4 + 1 + 0 + 1 + 4 = 10.$$

Thus,

$$s^2 = \frac{10}{5-1} = 2.5, \quad s = \sqrt{2.5} \approx 1.581.$$

Since $\sigma = \sqrt{4} = 2$,

$$\bar{x} = 10, \quad s^2 = 2.5, \quad s \approx 1.581.$$

Here $\bar{x} = \mu$, while $s^2 < \sigma^2$ and $s < \sigma$; this sample has less spread than the population model.

C7.2 – Local delivery time

$$n = 6, \quad \bar{x} = \frac{18 + 20 + 21 + 19 + 22 + 20}{6} = 20.$$

$$\sum (x - \bar{x})^2 = 4 + 0 + 1 + 1 + 4 + 0 = 10,$$

so

$$s^2 = \frac{10}{6-1} = 2, \quad s = \sqrt{2} \approx 1.414.$$

The theoretical variance is $\sigma^2 = 1.5^2 = 2.25$. Therefore,

$$\bar{x} = 20, \quad s^2 = 2, \quad s \approx 1.414.$$

The sample mean equals μ ; both measures of sample spread are slightly below their theoretical values 2.25 and 1.5.

C7.3 – Reusable cups

From the value–frequency table,

$$n = \sum f = 2 + 5 + 4 + 1 = 12,$$

$$\sum fx = 2(1) + 5(2) + 4(3) + 1(4) = 28,$$

$$\sum fx^2 = 2(1^2) + 5(2^2) + 4(3^2) + 1(4^2) = 74.$$

Hence

$$\bar{x} = \frac{28}{12} = \frac{7}{3} \approx 2.333,$$

$$s^2 = \frac{74 - 12(7/3)^2}{11} = \frac{26}{33} \approx 0.788,$$

$$s = \sqrt{\frac{26}{33}} \approx 0.888.$$

Since $\sigma = 1$,

$$n = 12, \quad \bar{x} \approx 2.333, \quad s^2 \approx 0.788, \quad s \approx 0.888.$$

All three sample statistics are below the corresponding theoretical values 2.5, 1, and 1.

C7.4 – Mobile-data use

$$n = 1 + 3 + 4 + 2 = 10,$$

$$\sum fx = 1(48) + 3(50) + 4(52) + 2(54) = 514,$$

$$\begin{aligned} \sum fx^2 &= 1(48^2) + 3(50^2) + 4(52^2) + 2(54^2) \\ &= 26452. \end{aligned}$$

Therefore,

$$\bar{x} = \frac{514}{10} = 51.4,$$

$$s^2 = \frac{26452 - 10(51.4)^2}{9} = 3.6,$$

$$s = \sqrt{3.6} \approx 1.897.$$

Thus

$$n = 10, \quad \bar{x} = 51.4, \quad s^2 = 3.6, \quad s \approx 1.897.$$

The mean is 0.6 below $\mu = 52$; the variance and standard deviation are also slightly below 4 and 2.

C7.5 – Commute time

The class midpoints of [10, 20), [20, 30), [30, 40), and [40, 50) are 15, 25, 35, 45. Thus,

$$n = 2 + 5 + 4 + 1 = 12,$$

$$\sum fx = 2(15) + 5(25) + 4(35) + 1(45) = 340,$$

$$\begin{aligned} \sum fx^2 &= 2(15^2) + 5(25^2) + 4(35^2) + 1(45^2) \\ &= 10500. \end{aligned}$$

Using the midpoints,

$$\bar{x} = \frac{340}{12} \approx 28.333,$$

$$s^2 = \frac{10500 - 12(340/12)^2}{11} = \frac{2600}{33} \approx 78.788,$$

$$s \approx \sqrt{78.788} \approx 8.876.$$

Therefore,

$$n = 12, \quad \bar{x} \approx 28.333, \quad s^2 \approx 78.788, \quad s \approx 8.876.$$

The theoretical variance is $\sigma^2 = 10^2 = 100$. The grouped-sample estimates are below $\mu = 30$, $\sigma^2 = 100$, and $\sigma = 10$.

C7.6 – Parcel mass

The class midpoints of $[0, 2)$, $[2, 4)$, $[4, 6)$, and $[6, 8)$ are 1, 3, 5, 7. Therefore,

$$n = 1 + 4 + 4 + 1 = 10,$$

$$\sum fx = 1(1) + 4(3) + 4(5) + 1(7) = 40,$$

$$\sum fx^2 = 1(1^2) + 4(3^2) + 4(5^2) + 1(7^2) = 186.$$

Thus,

$$\bar{x} = \frac{40}{10} = 4,$$

$$s^2 = \frac{186 - 10(4)^2}{9} = \frac{26}{9} \approx 2.889,$$

$$s = \sqrt{\frac{26}{9}} \approx 1.700.$$

Since $\sigma = \sqrt{3} \approx 1.732$,

$$n = 10, \quad \bar{x} = 4, \quad s^2 \approx 2.889, \quad s \approx 1.700.$$

The estimated mean equals μ ; the estimated variance and standard deviation are close to, but below, their theoretical values.

C7.7 – Package preparation time

There are three individual observations and seven observations in the value–frequency table. Therefore,

$$n = 3 + (2 + 3 + 2) = 10.$$

Combining both representations,

$$\begin{aligned} \sum x &= (8 + 10 + 12) + 2(9) + 3(10) + 2(11) \\ &= 100. \end{aligned}$$

Thus,

$$\bar{x} = \frac{100}{10} = 10.$$

For the sum of squares,

$$\begin{aligned} \sum x^2 &= (8^2 + 10^2 + 12^2) + 2(9^2) + 3(10^2) + 2(11^2) \\ &= 1012. \end{aligned}$$

Therefore,

$$\begin{aligned} s^2 &= \frac{1012 - 10(10)^2}{10 - 1} \\ &= \frac{12}{9} \approx 1.333, \end{aligned}$$

and

$$s = \sqrt{\frac{4}{3}} \approx 1.155.$$

Hence,

$$n = 10, \quad \bar{x} = 10, \quad s^2 \approx 1.333, \quad s \approx 1.155.$$

The sample mean equals the theoretical mean $\mu = 10$.

The sample variance and standard deviation are below the theoretical values $\sigma^2 = 2$ and $\sigma = \sqrt{2} \approx 1.414$.

C7.8 – Customer service time

The grouped intervals have midpoints

$$15, \quad 25, \quad 35, \quad 45.$$

There are four individual observations and six grouped observations, so

$$n = 4 + (1 + 2 + 2 + 1) = 10.$$

Combining the individual observations with the grouped-data estimates,

$$\begin{aligned} \sum x &= (22 + 28 + 32 + 38) + 1(15) + 2(25) + 2(35) + 1(45) \\ &= 300. \end{aligned}$$

Therefore,

$$\bar{x} = \frac{300}{10} = 30.$$

Similarly,

$$\begin{aligned} \sum x^2 &= (22^2 + 28^2 + 32^2 + 38^2) \\ &\quad + 1(15^2) + 2(25^2) + 2(35^2) + 1(45^2) \\ &= 9686. \end{aligned}$$

Hence,

$$\begin{aligned} s^2 &= \frac{9686 - 10(30)^2}{10 - 1} \\ &= \frac{686}{9} \approx 76.222, \end{aligned}$$

and

$$s = \sqrt{\frac{686}{9}} \approx 8.731.$$

Therefore,

$$n = 10, \quad \bar{x} = 30, \quad s^2 \approx 76.222, \quad s \approx 8.731.$$

The estimated sample mean equals $\mu = 30$. The estimated standard deviation is slightly below the theoretical value $\sigma = 9$. Because part of the sample is grouped, the statistics obtained using class midpoints are estimates.